

12 (a) takes all the simultaneous digits for one number
and 'sends' them one after another (along the transmission line)

B1
B1 [2]

(b) (i) 0111

A1 [1]

(ii) 0110

A1 [1]

(c) levels shown

t	0	0.2	0.4	0.6	0.8	1.0	1.2
	0	8	7	15	6	5	8

(-1 for each error or omission)
correct basic shape of graph i.e. series of steps
with levels staying constant during correct time intervals
(vertical lines in steps do not need to be shown)

A2
M1
A1 [4]

(d) increasing number of bits reduces step height
increasing sampling frequency reduces step depth / width
reproduction of signal is more exact

M1
M1
A1 [3]